



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2024/2025

**FIRST YEAR FIRST SEMESTER EXAMINATION FOR THE
DEGREE OF BACHELOR OF SCIENCE IN COMPUTER
TECHNOLOGY, BACHELOR OF COMPUTER SCIENCE
AND BACHELOR OF MATHEMATICS & COMPUTER
SCIENCE**

MAIN CAMPUS

CCS 117: FUNDAMENTALS OF COMPUTING

Date: 21st January, 2025

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Do not write anywhere on this Question paper
- Answer Question ONE and any other TWO

QUESTION ONE [30 marks]

- a) With the aid of an illustrated sketch explain the steps in information processing cycle of a computer system. Give two examples of the components of the system involved at each stage of the cycle. [4 mks]
- b) Draw a simple model of a computer system illustrating the relationship among computer hardware, System software and application software. Give an example of each type of component. [4 mks]
- c) With regard to Windows 7 operating system, explain
- i) The meaning of the following power modes Restart, Sleep, Hibernate and Shutdown. [2 mks]
 - ii) The difference between cold boot and warm boot. [2 mks]
 - iii) The booting up procedure of the operating system [2 mks]
- d) State and explain the various functions of a computer operating system, giving three examples of popular desktop operating systems. [3 marks]
- e) Explain the operational features of the following computer input and output devices: keyboard, mouse, touchpad, CRT monitor, laserjet printer and speakers. [6 mks]
- f) State the typical computer security risks in modern organizations. [2 mks]
- g) A user creates a word document and sends this as an email to a remote user on the Internet. Explain how a virus attack can spread through this email message on the Internet. [2 mks]
- h) Explain the role of programming languages in the use of computer systems. Give three examples of popular programming languages. [3 mks]

QUESTION TWO [20 marks]

- a) There are two schools of thought relating to development and advancement of computer systems; the first school of thought states that "*Business enterprise activities influence development and advancement of Computers systems*" while the second school of thought argues that "*Development and advancement of computer Systems technology influences and shapes business activities*". Discuss citing appropriate examples, the validity of these schools of thought. [6 Marks]
- b) Draw the block diagram of the basic hardware components of a computer systems and state the function of each component. [5 marks]
- c) Explain the Fetch, Decode and Execute cycle of the computer CPU, stating which components of the CPU are involved for which cycle. [5 marks]
- d) Discuss any **FOUR** factors that would be considered when purchasing a visual display unit (monitor) for a computer system and how each of the factors impacts display of the monitor. [4 Marks]

QUESTION THREE [20 Marks]

- a) The first electronic digital computers were developed between 1940 and 1945. Thereafter computers started being studied in groups called generations. Explain the characteristics of the following generations of computers.
- | | | |
|-------|-------------------|-----------|
| (i) | Second Generation | [3 marks] |
| (ii) | Fourth Generation | [3 marks] |
| (iii) | Fifth Generation | [3 marks] |
- b) State and explain THREE features of the third generation languages that make them useful when developing an Information Systems. [3 marks]
- c) One of the functions of an operating system is to manage processes. Discuss this with special reference to
- | | | |
|-------|------------------------|-----------|
| (i) | Multiprocessing , | [2marks] |
| (ii) | Multiprogramming. | [2 marks] |
| (iii) | Multitasking | [2marks] |
| (iv) | Non-emptive processing | [2marks] |

QUESTION FOUR [20 marks]

- a) Perform the following number conversions
- Decimal to binary: 192
 - Binary to decimal: 11111100
 - Hexadecimal to decimal: 1234
 - Decimal to Hexadecimal: 512
 - Hexadecimal to binary: ABCD
 - Binary to Hexadecimal: 110100101
- [6 mks]
- b) With regard to computer input and output subsystem
- What is meant by USB and what are its advantages over serial and parallel ports? [3 mks]
 - Explain, using a simple sketch, the role of an expansion bus. Give two examples of popular computer bus standards. [5 mks]
- c) A computer video standard, SXGA, specifies a resolution of 1280 x 1024 and a 24-bit colour system. If the video adapter card supports a refresh rate of 72 Hz, calculate the minimum data rate that should be supported by the AGP bus, which connects to the monitor. [6 mks]

QUESTION FIVE [20 marks]

- a) With regard to computer networks, explain the meanings of the terms
- Intranet, Extranet and Internet.
 - LAN, MAN and WAN
- [6 mks]
- b) Overheating in computers may cause malfunctioning. Describe in details the two approaches that manufactures use to manage overheating challenges. [6 mks]
- c) Explain the difference between an operating system and utility programs. Give two popular examples of each type of system software. [3 marks]
- d) With respect to Word 2007, explain
- How to perform the following editing operations: basic selection of a text and how to copy items
 - How to insert symbols into your document
 - How to perform the following formatting operations Font style, size, and shape (Bold, Italic and Underline)
 - How to convert a word document to and save it as pdf document
 - How to print an open document
- [5 mks]